

CSE 309 Theory of Automata

School of Mathematics and Computer Science
Institute of Business Administration

Spring 2026

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Offices: Room 210, Second Floor, North Tabba Building, Main Campus

Office Hours: Mon/Wed 1:30pm – 3:30pm, available online via class Discord or email

Class meeting times: Mon/Wed (10:00 – 11:15 am)

Mon/Wed (11:30 – 12:45 pm)

Class Location: Auditorium, IBA Main Campus

Class website: LMS

Discord: Discord Server Link for Class Discussions

Assignment submission: Gradescope (Class code: ZJ3RB3)

Tutorial meeting times:

Abdul Wasay, TBA

Zuha, TBA

Imad, TBA

Itbaan, TBA

Ali, TBA

1 Course Objectives

This course is about the beautiful contributions and insights that the field of computer science has afforded us. We shall specifically focus on the interplay between notions of logical proof and how they link with limits on computation; which in turn has implications on what can be computed and how efficiently it can be computed in the physical world. The perspective we gain by looking at our world through the computational lens is perhaps the main contribution of theoretical computer science to the human endeavor.

The course starts from basic math preliminaries, works its way through historical breakthrough results in the field and then branches off into an investigation of fundamental challenges at the frontiers of theoretical computer science. The course provides opportunities for both:

- (i) practice in developing a rigorous mathematical argument, and

- (ii) practice in developing a clear intuitive explanation of a complex argument.

During Introduction to Programming and Discrete Math students learn a new language, i.e., Discrete Math. They obtain some practice in reading, writing, speaking and listening to this new language. In a human languages class, a logical progression is to read and understand the existing classic literature in the language. Similarly, we will engage ourselves with some of the classics in theoretical computer science.

The course content is spread over four distinct modules. The first deals with setting up the mathematical framework and language for our investigation. The second deals what can be computed in principle: *computability*, the third deals with how efficiently we can compute: *complexity* and finally we branch off into an investigation of fundamental challenges at the frontiers of theoretical computer science.

2 Learning Outcomes

By the end of this course, students will be able to

1. model computational tasks in an appropriate computational framework. (PLO3,4 - Analysis / Design)
2. determine limitations and advantages of different computational frameworks. (PLO2,3 - Knowledge / Analysis)
3. identify restrictions on computational models that emerge from physical limitations. (PLO2 - Knowledge)
4. construct arguments to show that certain computational questions are uncomputable. (PLO3,4,10 - Analysis / Design / Life-long)
5. identify commonly used classes that occur in the study of complexity theory. (PLO2 - Knowledge)
6. reduce computationally intractable problems to the class NP-Complete. (PLO2,3,10 - Knowledge / Analysis / Life-long)
7. be introduced to ideas from the frontiers of theoretical computer science. (PLO10)

3 Course Texts

The main text for the course is

- *Introduction to the Theory of Computation* by Michael Sipser.

Students interested in a more advanced treatment of the material may refer to the following texts.

- *Computational Complexity: A Modern Approach* by Sanjeev Arora and Boaz Barak.

- *Mathematics and Computation* by Avi Wigderson.
- *The Emperor's New Mind* by Roger Penrose.
- *Quantum Computing Since Democritus* by Scott Aaronson.
- *Feynman Lectures on Computation* by Richard P. Feynman.

If necessary, additional readings will be made available online.

4 Grading Policy

Students will be graded on the following basis.

06%	Open Problem Exploration
07%	AI Assisted Exercise Review
24%	4 Problem Sets, 6% each
08%	Atleast 4 quizzes, 2% each
25%	Midterm Exam
30%	Final Exam
+ up to 5%	Participation bonus, ungraded exercises, contribution in class, attendance,utilization of office hours. These are distributed entirely at discretion of the instructor on a need and merit basis.

4.1 Problem Sets

The main tool for learning the concepts we discuss in class are the problem sets. The ‘personal struggle’ you engage in with these problem sets will allow you to develop the skills necessary for success as a theoretical computer scientist. ***Always*** spend some time thinking about these problems on your own before asking your favorite AI model for solutions, asking for hints etc. Learning the material happens when you are working on problems rather than looking up complete solutions.

You are welcome to collaborate on problem sets, provided that (1) you write up your solutions individually, and (2) you clearly cite the names of all collaborators and sources. Failure to do so will result in zero credit. An additional key requirement is that you should be able to explain what you submit. Inability to do so will result again in zero credit. Your solution arguments should be clearly written up (L^AT_EX is strongly encouraged).

Problem set submission is only through the site gradescope. Use the entry code ‘ZJ3RB3’ to add yourselves to the course. It is strongly recommended to use provided L^AT_EX template for writing up your problem set solutions.

Problem set 1 (Out Week 2)	Due Week 4, Monday, 9 Feb 11:59 pm
Problem set 2 (Out Week 4)	Due Week 6, Monday, 23 Feb 11:59 pm
Problem set 3 (Out Week 10)	Due Week 13, Monday, 13 Apr 11:59 pm
Problem set 4 (Out Week 13)	Due Week 15, Monday, 27 Apr 11:59 pm

4.2 Exercises and Quizzes

One of the main tools used in the class to ensure consistent learning are the weekly exercises. The exercises will cover relatively simple questions that deal with the content covered in the week's classes. This semester we will also experiment with an AI assisted model for the exercises. Details will be shared in class. Each tutorial section will review the exercises for that week so it is an excellent opportunity to get your questions answered. It is strongly recommended that you attempt the exercise questions before coming to tutorials.

We will have at least four quizzes during the semester. Schedule is on LMS.

5 Late Policy

No late solutions will be accepted and no make-up exams will be given. If you have a valid medical excuse, the percentage of your grade corresponding to the missed work will be shifted to the final exam. Valid excuses require supporting documentation from a doctor.

6 Attendance Policy

IBA attendance policy applies.

7 Academic Integrity

Each student in this course is expected to abide by the IBA Code of Conduct.

Scholastic dishonesty shall be considered a serious violation of these rules and regulations and is subject to strict disciplinary action as prescribed by IBA regulations and policies. Scholastic dishonesty includes, but is not limited to, cheating on exams, plagiarism on assignments, and collusion.

- **PLAGIARISM:** Plagiarism is the act of taking the work created by another person or entity and presenting it as one's own for the purpose of personal gain or of obtaining academic credit. Plagiarism includes the submission of or incorporation of the work of others without acknowledging its provenance or giving due credit according to established academic practices. This includes the submission of material that has been appropriated, bought, received as a gift, downloaded, or obtained by any other means. Students must not, unless they have been granted permission from all faculty members concerned, submit the same assignment or project for academic credit for different courses.
- **CHEATING:** The term cheating shall refer to the use of or obtaining of unauthorized information in order to obtain personal benefit or academic credit.
- **COLLUSION:** Collusion is the act of providing unauthorized assistance to one or more person or of not taking the appropriate precautions against doing so.

Any student violating academic integrity a second time in this course will receive a failing grade for the course, and additional disciplinary sanctions may be administered.